

# Question Paper

Exam Date & Time: 24-Dec-2025 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: ECE 2126/ECE\_2126 ANALOG CIRCUITS

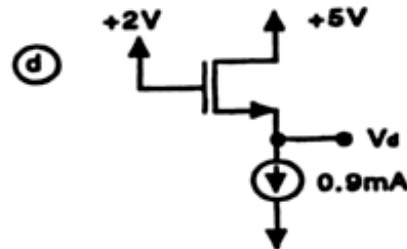
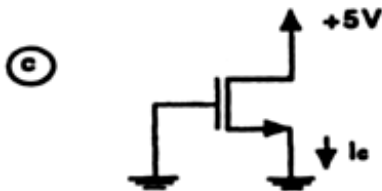
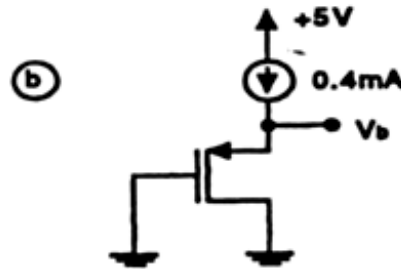
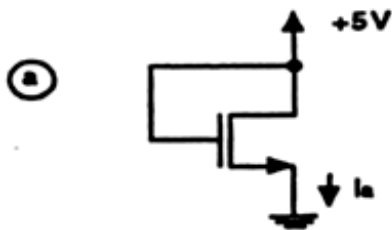
(SPL: Electronics Engineering (VLSI Design and Technology))

Marks: 50

Duration: 180 mins.

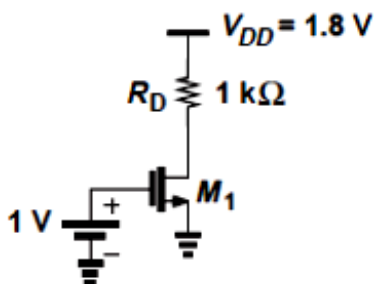
Answer all the questions.

- 1A) For the following circuits employing enhancement MOSFETs, for which  $|V_t| = 2V$  and  $\mu_n C_{ox} = 20 \mu A/V^2$ ,  $W = 20 \mu m$ ,  $L = 2 \mu m$ , find the labelled voltages and currents. (4)



Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) Assume  $\lambda = 0$ , compute  $W/L$  of  $M_1$  in the below Figure such that the device operates at the edge of saturation. Also, using the value of  $W/L$  found in the problem, explain what happens if the gate oxide thickness is doubled due to a manufacturing error. (3)

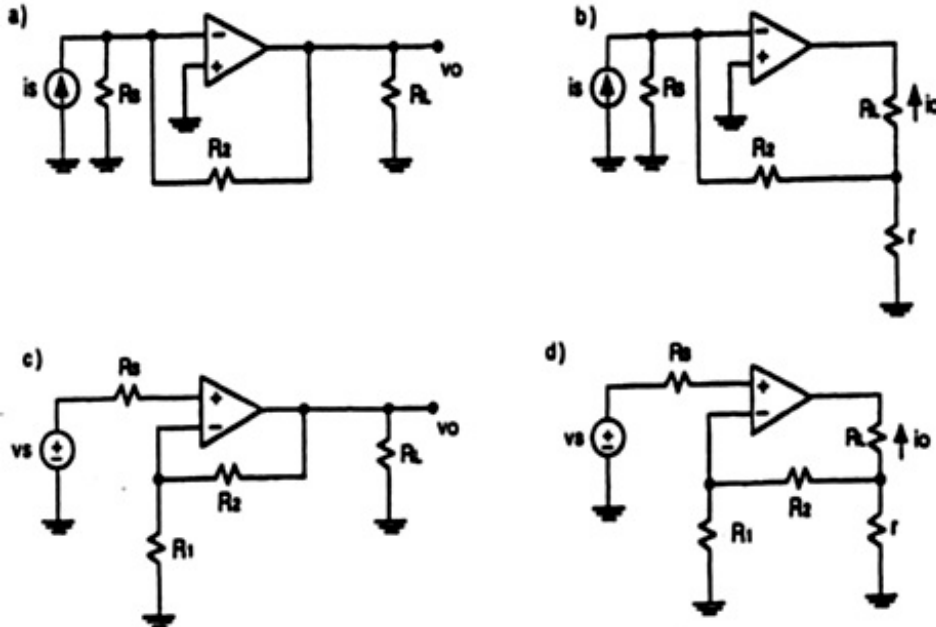


Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Explain the basic block diagram of an operational amplifier with a neat diagram and plot its transfer characteristics. Also, explain the characteristics of an ideal op-amp with proper definitions. (3)

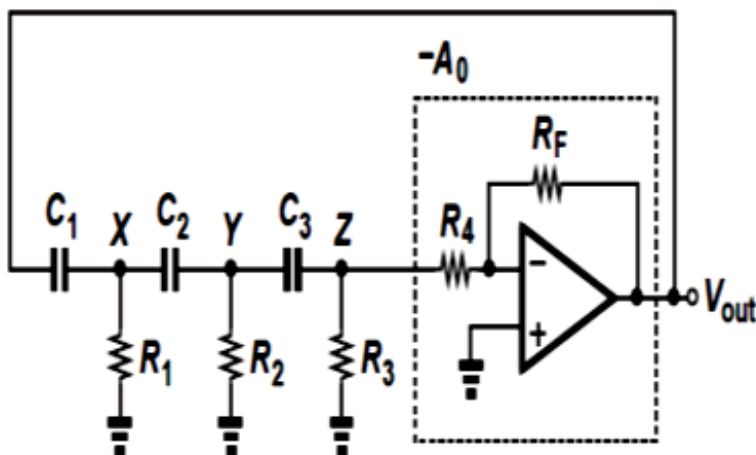
Course Outcome: "CO1" / Cognitive Level: "Remember"

- 2A) Characterize each of the following amplifiers by feedback type. As well for each, find  $\beta$  in terms of the labelled components. In all cases, assume that the op-amp is ideal. (4)



Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) Design the phase shift oscillator of Figure shown below for a frequency of 10 MHz, assuming an ideal opamp. Assume  $C_1 = C_2 = C_3 = 1\text{ nF}$ . (3)

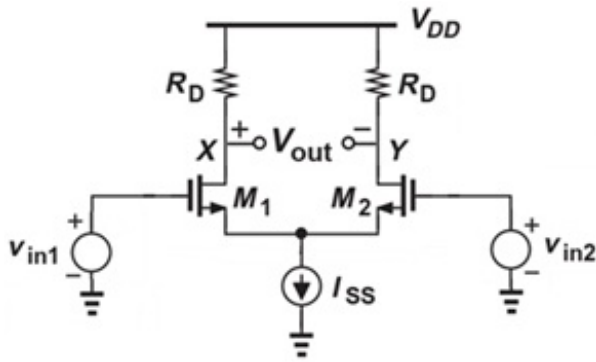


Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Explain the concept of Voltage-Voltage feedback with a neat diagram and derive the expressions for effective input impedance, output impedance, and gain. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

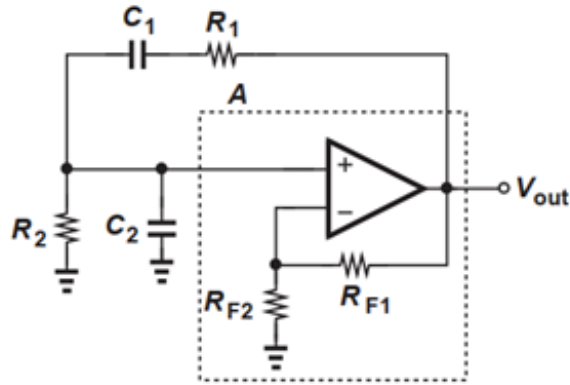
- 3A) Design the MOS differential amplifier shown in figure, for operating overdrive voltage of 0.25 V and to provide a transconductance of 1 mA/V. Consider  $V_{TH} = 0.5\text{ V}$  and  $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$ . (4)



Course Outcome: "CO2" / Cognitive Level: "Application"

3B)

(3)

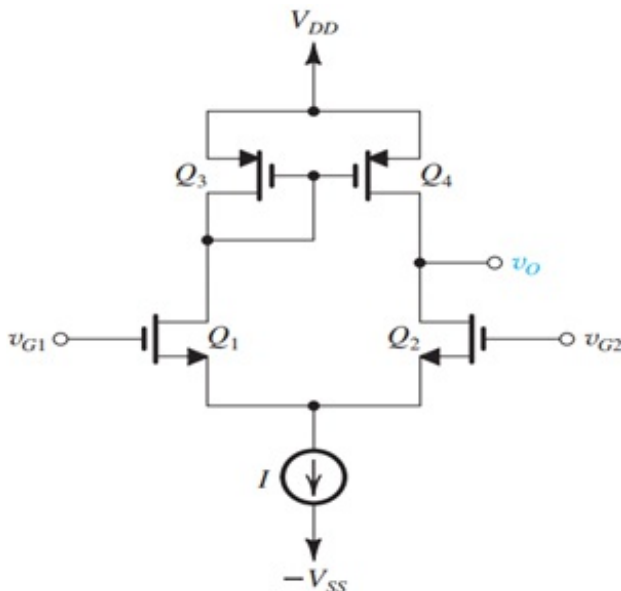


Design the Wein-bridge oscillator shown in the figure for a frequency of 2 MHz, assuming an ideal Op-amp. Let  $C_1 = C_2 = 220 \text{ pF}$ . Determine  $R_1 = R_2$  and feedback resistors  $R_{F1}, R_{F2}$  so the circuit will sustain oscillation.

Course Outcome: "CO2" / Cognitive Level: "Application"

3C)

(3)

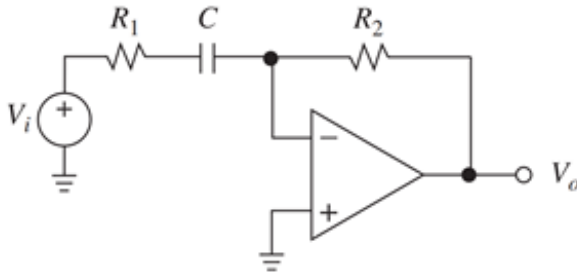


A current-mirror-loaded NMOS differential pair has the following parameters:  $\mu_n C_{ox} = 100 \text{ } \mu\text{A/V}^2$ ,  $\mu_p C_{ox} = 50 \text{ } \mu\text{A/V}^2$ ,  $(W/L)_n = 40/0.5$ ,  $(W/L)_p = 80/0.5$ ,  $I_{SS} = 0.8 \text{ mA}$ ,  $\lambda_n = 0.05 \text{ V}^{-1}$ , and  $\lambda_p = 0.1 \text{ V}^{-1}$ .

Calculate

- Transconductance  $g_m$  of input transistors  $Q_1$  and  $Q_2$ .
- Output resistance  $r_{o2}$  and  $r_{o4}$
- Differential gain  $A_d$

4A) (4)



A high-pass active filter with gain is implemented as shown in figure. The filter has the following measured characteristics:

- At  $f = 50 \text{ Hz}$ :  $|H| = 0.5 \text{ V/V}$ ,  $\angle H = +45^\circ$
- At  $f = 10 \text{ kHz}$ :  $|H| = 12 \text{ V/V}$ ,  $\angle H = +5^\circ$

(a) Analyze the circuit to determine the 3-dB cutoff frequency ( $f_0$ ) and high-frequency gain ( $H_0$ ).

(b) If  $R_1 = 47 \text{ k}\Omega$  and  $C = 68 \text{ nF}$ , calculate  $R_2$  and verify whether still the circuit works as HPF for the  $H_0$  and  $f_0$  obtained in the part (a). If not suggest the values changes in the C and  $R_2$  components.

Course Outcome: "CO4" / Cognitive Level: "Application"

4B) Using standard 5% resistor values, design a two stage circuit that produces the output voltage  $V_o = 8V_1 + 12V_2 - 4V_3$ . Provide the complete circuit diagram and a detailed step-by-step solution justifying your component choices. The input impedance of 1st stage is  $10 \text{ k}\Omega$ . Hint: You can use unity gain inverting amplifier and the 1st stage with followed by difference amplifier in the 2nd stage. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

4C) What is an Opamp window detector? Explain the working of it with relevant circuit diagram and its voltage transfer characteristics. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

5A) The standard astable multivibrator circuit using a 555 timer cannot achieve a duty cycle of exactly 50%. Propose and explain a circuit (with potentiometer) modification to achieve i) duty cycle less than 50% ii) duty cycle more than 50% (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

5B) Design a simple automatic light control system using a 555 timer for a staircase. When a person presses a push button, a light bulb should turn ON and remain ON for 2 minutes, after which it should automatically turn OFF. The system uses a 9V battery. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) Explain the advantages of using a current-mirror-loaded differential amplifier over a traditional resistor loaded differential pair for single-ended output applications. Derive the differential voltage gain for the current-mirror-loaded configuration. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

-----End-----

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-MIT BLR-MAKEUP EXAM-2025-2026

ECE\_2126:Analog Circuits

| # | Course Outcome | Description | Marks |
|---|----------------|-------------|-------|
|---|----------------|-------------|-------|

|              |   |      |           |
|--------------|---|------|-----------|
| 1            | 1 | CO 1 | 10        |
| 2            | 2 | CO 2 | 17        |
| 3            | 3 | CO 3 | 6         |
| 4            | 4 | CO 4 | 10        |
| 5            | 5 | CO 5 | 7         |
| <b>Total</b> |   |      | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-MIT BLR-MAKEUP EXAM-2025-2026      ECE\_2126:Analog Circuits

| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 1               | 3(6%)     |
| 2     | Understand  | 3               | 9(18%)    |
| 3     | Application | 9               | 30(60%)   |
| 4     | Analysis    | 2               | 8(16%)    |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 15              | 50 (100%) |